AssemblyImpact AI — P3 Engineering Change Case Worklist: Implementation, Testing & Evidence Report

SAP Fiori elements List Report / Object Page on the EngineeringReview CAP service, generated with SAP Project Accelerator and adapted

Repository	Naman028/assemblyimpact-ai
Branch	feat/p3-case-worklist
Code head commit	377e46aa5b712487fa63777f9c0ebe38e00271bf
Base (main)	a0068c6050ca09efae12deb3eacc26a6a4d2539d
Environment	SAP BTP trial — SAP Business Application Studio (dev space ws-lecln)
Pull request	not yet created
CI	GitHub Actions: not yet available when this report was generated
Report date	2026-11-09
Prepared by	Claude (Anthropic) for Naman Patel — AI-generated; student review pending

All demonstration records in this report are synthetic (fictional NovaMotion Manufacturing). They are not SAP S/4HANA data. Screenshots are unedited captures of the actual BAS session.

1. Executive summary

Objective. Begin P3: generate a Fiori elements Case Worklist with SAP Project Accelerator in an isolated sandbox, run the generated UI, and connect a corrected version to the existing read-only EngineeringReview OData V4 service.

Status. Implemented and verified in a running browser session in SAP Business Application Studio (BAS). Branch `feat/p3-case-worklist` (3 commits on main a0068c6). GitHub Actions: not yet available when this report was generated. Draft pull request: not yet created. Not merged, not deployed.

- Project Accelerator ran in the separate sandbox `pa-sandbox/case-worklist`; its original 44 files are preserved unchanged (sandbox commit 9910c02, copy in `generated/`).
- The generated app was run in the browser with its own mock data (P3-08, P3-09). It showed invented values and draft editing, so only its page structure was reused.
- The implemented app `app/case-worklist` shows Review Cases -> Review Case -> Impact Run -> Finding from EngineeringReview, with the Run Impact action.
- Browser checks: plant isolation, 403 for viewers on Run Impact, analyst run reproducing the oracle (+1,200.00 EUR, 2 short, 7 findings), cross-plant deep link 404.
- Three defects found only in the browser were fixed: UI5 1.120 startup error, zero-height app container, unsorted section tables.
- npm test 40/40 and lint pass locally and in BAS; no new dependencies.

Not claimed. The Project Accelerator Page Map (P3-05) is a design-time view, not a working UI; the working UI is the browser-rendered application in P3-08 to P3-20. No BTP deployment, no S/4HANA data, no Joule, no P4 work. All records are synthetic.

2. Starting state and inputs

Item	Value
GitHub baseline	main = a0068c6050ca (M04 Slice 1b, PR #4). BAS fast-forwarded; 35/35 tests (P3-01).
Backend	CAP OData V4 EngineeringReview: ReviewCases, ImpactRuns, Findings, ImpactMaterialLines, ImpactComparisonLines; read-only plus bound action runImpact (ChangeAnalyst, plant-restricted).
Seed data	SEED-M04-01 (synthetic): EC-000473 / COOL-100 plant 1000 and mirror EC-200473 plant 2000. No impact runs are seeded.
Design input	Approved layout mockup (illustrative values) and requirements AssemblyImpact_Case_Worklist_requirements.md (sha256 021aa8e1...).
Runtime (BAS)	Node v24.21.0, @sap/cds 10.2.2, cds-compiler 7.2.1, @cap-js/sqlite 3.1.2, @sap/cds-fiori 2.3.0; SAPUI5 1.149.0 (CDN).

3. Project Accelerator generation (sandbox)

Project Accelerator (SAP Fiori tools, CAP Accelerator flow) was given the requirements text and the layout image (P3-02 to P3-04). The output was committed in the sandbox as 9910c02 and archived (`pa-original-output.tgz`, sha256 e3af883acbae...). The original files are copied unchanged to `docs/evidence/p3-case-worklist/generated/`; the archive itself is not committed.

Generated part	Content	Used?
Data model	Own namespace assemblyImpactCaseWorklist: ReviewCases, Assemblies, Findings, MaterialImpacts, DecisionHistories + 6 code lists	No (D-P3-2)
Service	assemblyImpactCaseWorklistSrv at /odata/v4/assembly-impact-case-worklist-srv/, @odata.draft.enabled	No
Sample data	CSV rows invented by the generator (10 cases, findings, materials)	No
Fiori app	List Report -> Object Page -> Findings Object Page; minUI5Version 1.149.0	Page structure only

Generated part	Content	Used?
Metadata	.fiori-ai: log.txt, model.json, user-input.txt, userImage.png	Evidence only

4. Generated UI preview and design comparison

The generated application was started in the sandbox and opened in the browser (P3-08, P3-09). It renders, but with the generator's own invented records and draft editing (Create, Edit, Delete, Editing Status), which conflicts with the read-only, role- and plant-checked EngineeringReview service.

Mockup / requirement	Project Accelerator output	Implemented	Reason
Filters: Assembly, Plant, Revision, Status, Source	Filters on generated entity + Editing Status	assemblyCode, plant, toRevision, status, sourceType	same intent on existing service
Columns up to Review Status	Columns on generated entity	9 columns; findings / short / cost delta from latestRun	figures come from immutable runImpact snapshots
Status colours	Code-list criticality	$edmJson If/Eq criticality on status	no new backend field
Create Change, Export	Create/Delete (draft)	Not implemented	read-only service (D-P3-5)
Case: General information	Object Page section	FieldGroup General + Latest Impact	-
Findings tab on the case	/ReviewCases/Findings	Impact Run page -> Findings -> Finding page	findings stored per run (D-P3-3)
Material Impact tab	MaterialImpacts table	Impact Run page: Material Impact + BOM Comparison	per run
Decision History tab	DecisionHistories entity	Not shown	no backend entity yet
Run impact	Not generated	runImpact action on list and case	server-enforced (D-P3-4)
Draft editing	@odata.draft.enabled	None	service read-only by design

5. Implementation record

Unit	Files	What	Decision
U1 App descriptor	app/case-worklist/webapp/manifest.json, Component.js, index.html, i18n	dataSource /odata/v4/engineering-review/; routes ReviewCasesList, ReviewCasesObjectPage, ImpactRunsObjectPage, FindingsObjectPage; no draft; UI5 1.149.0 bootstrap.	D-P3-2, D-P3-3, D-P3-6
U2 Annotations	app/case-worklist/annotations.cds	SelectionFields, LineItem with criticality and runImpact, HeaderInfo, FieldGroups, Facets, PresentationVariants on EngineeringReview entities.	D-P3-4, D-P3-7
U3 Tests	test/case-worklist-app.test.js, package.json	5 tests: manifest-model consistency, annotation contract, static serving, list/action authorization, navigation paths and plant isolation.	quality gate
U4 Documentation	app/case-worklist/README.md, readme.md	Origin, read-only scope, how to run with development users; synthetic-data notice.	traceability

Commit	Message
77730307a1ab	feat(p3): add Engineering Change Case Worklist Fiori elements app
af43aafe863e	fix(p3): correct standalone UI rendering found in BAS preview
377e46aa5b71	fix(p3): apply table sort orders on Object Page sections

All three commits: author and committer Naman Patel <108478447+Naman028@users.noreply.github.com>, no co-author trailer. The decision log is in `decisions.md`.

6. Corrections found in the running UI

#	Observed in the browser	Cause	Fix	Evidence
1	Blank page; console: storeInnerAppStateAsync is not a function	UI5 1.120.0 too old for the generated app pattern	Bootstrap SAPUI5 1.149.0 (af43aaf)	P3-10 onward
2	Still blank, no error	Component container had height 0	Full height for #container-uiarea (af43aaf)	P3-10 onward
3	Material Impact and BOM rows unsorted	Section facets targeted @UI.LineItem, so the PresentationVariant added in af43aaf was ignored	Facets target nav/@UI.PresentationVariant; test forbids LineItem facet targets (377e46a)	P3-18 -> P3-19, P3-20

Note on defect 3. The browser had cached the older $metadata, so screenshots P3-14/P3-15 (taken after af43aaf) show the unsorted state as well. After forcing a metadata reload, the table bindings were checked in the page: findings sorted by [type, material], material lines by [material], comparison lines by [itemNo]. Commit af43aaf's message overstated its effect on sorting; 377e46a is the effective fix.

Other observation. One cold deep-link load right after switching development users showed a transient "Not Found" dialog; it did not recur on reload or in-app navigation. Logged as an open observation.

7. Testing and verification

Command	Local	BAS
npm test	tests 40, pass 40, fail 0	tests 40, pass 40, fail 0 (after 377e46a)
npm run lint (eslint srv test --max-warnings=0)	exit 0	exit 0
Dependencies	package-lock.json unchanged	-

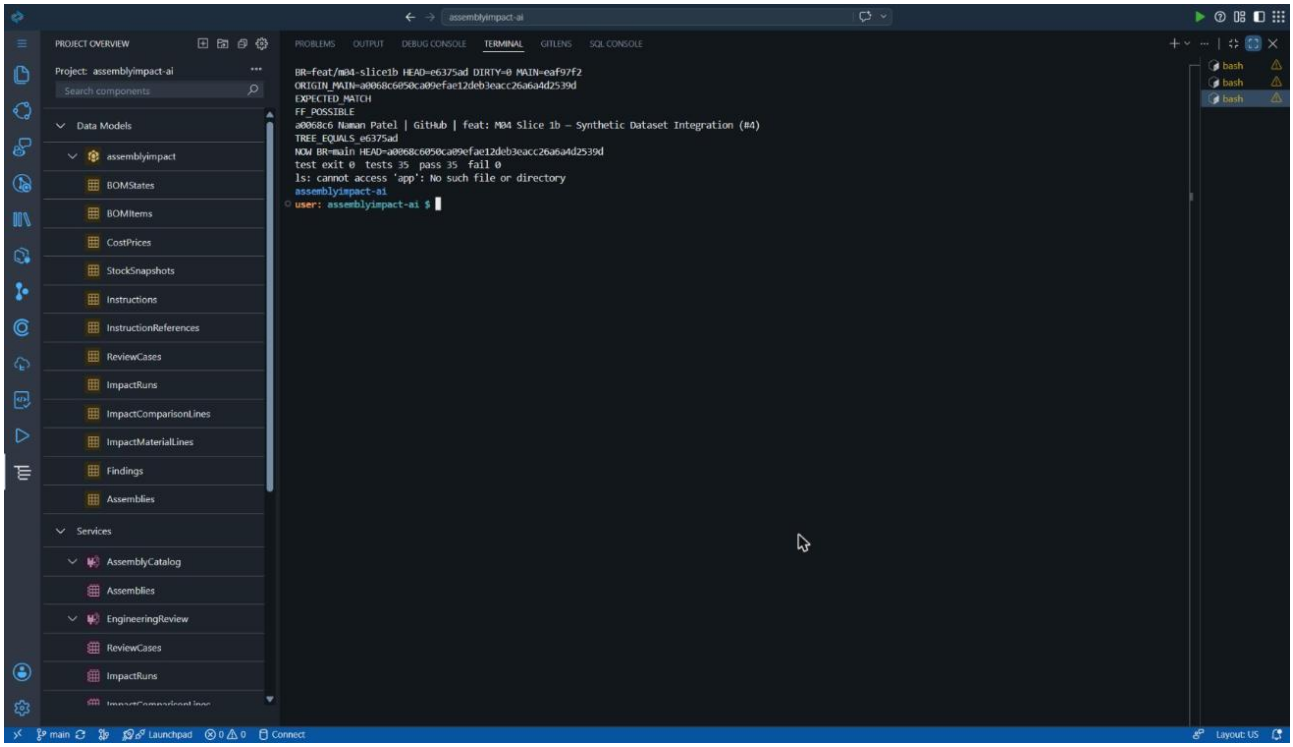
New test (test/case-worklist-app.test.js)	Checks	Result
manifest targets EngineeringReview	dataSource, entity sets and navigation paths resolve against the CDS model; no draft	pass
annotation contract	exact columns and filters, facets, criticality, sort orders, no LineItem facet targets, $metadata carries UI annotations	pass
static serving	index.html, manifest, Component, i18n served; full-height container; UI5 1.149.0 bootstrap	pass
list and action	plant-1000 user sees 1 case, plant-2000 user only EC-200473; runImpact 200 analyst / 403 viewer / 404 other plant; columns 1 / 2 / 1200	pass
navigation paths	case -> runs -> 7 findings / SHORT BRKT-01 50, FAN-B 20; cross-plant paths 404 or empty; anonymous 401	pass

Browser check (BAS preview, development profile)	Expected	Observed	Evidence
Viewer opens worklist	1 case, no impact figures yet	as expected	P3-10
Viewer presses Run Impact	403 message	403 Forbidden shown	P3-11
Analyst runs impact 2026-11-08	1 / 2 / 1,200.00	1 / 2 / 1,200.00	P3-12
Case and run pages	56.00 -> 68.00, 21.43 %, 7 findings, 2 SHORT	as expected	P3-13 to P3-15
Plant-2000 user	only EC-200473	as expected	P3-16
Cross-plant deep link	Not Found	Not Found	P3-17
Section sorting	material / item / type order	unsorted before, sorted after 377e46a	P3-18 to P3-20

8. Screenshot evidence

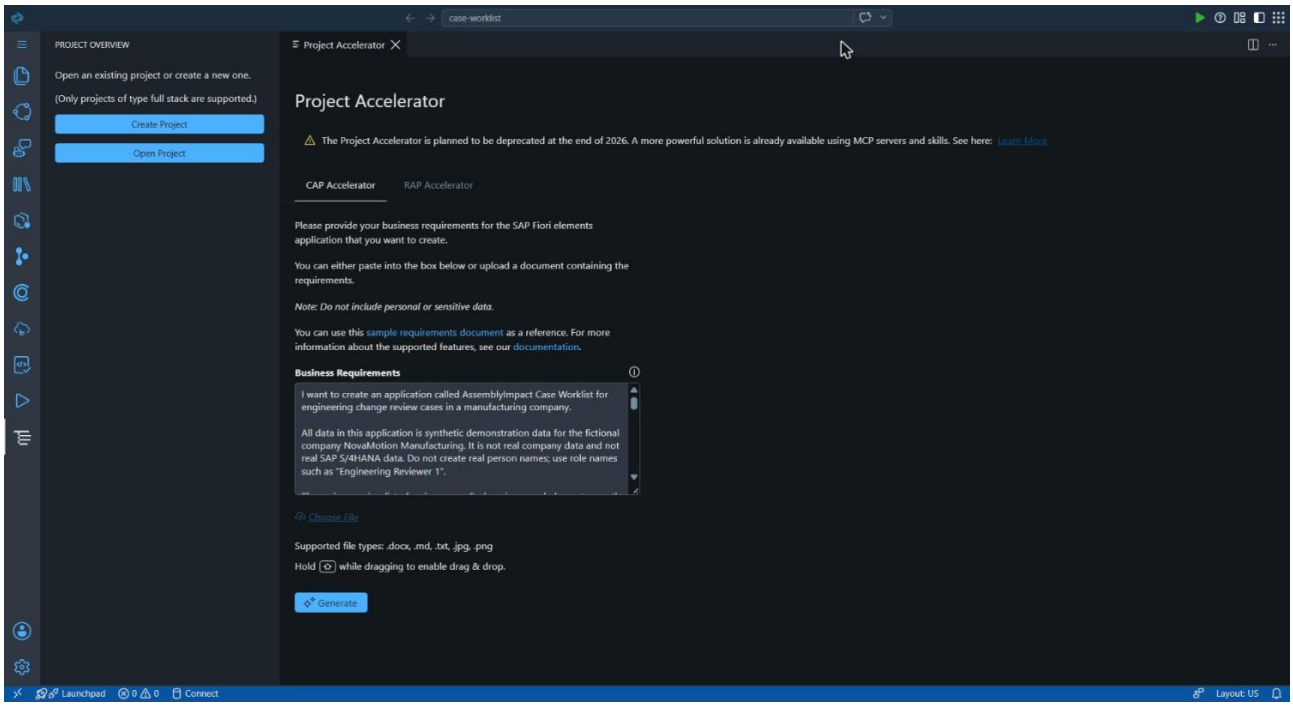
Original, unedited captures of the browser in the BAS session, taken with Claude in Chrome. Times are UTC. No secrets are visible; development passwords are public mock values and are not shown.

P3-01 — BAS synchronized to main and baseline regression



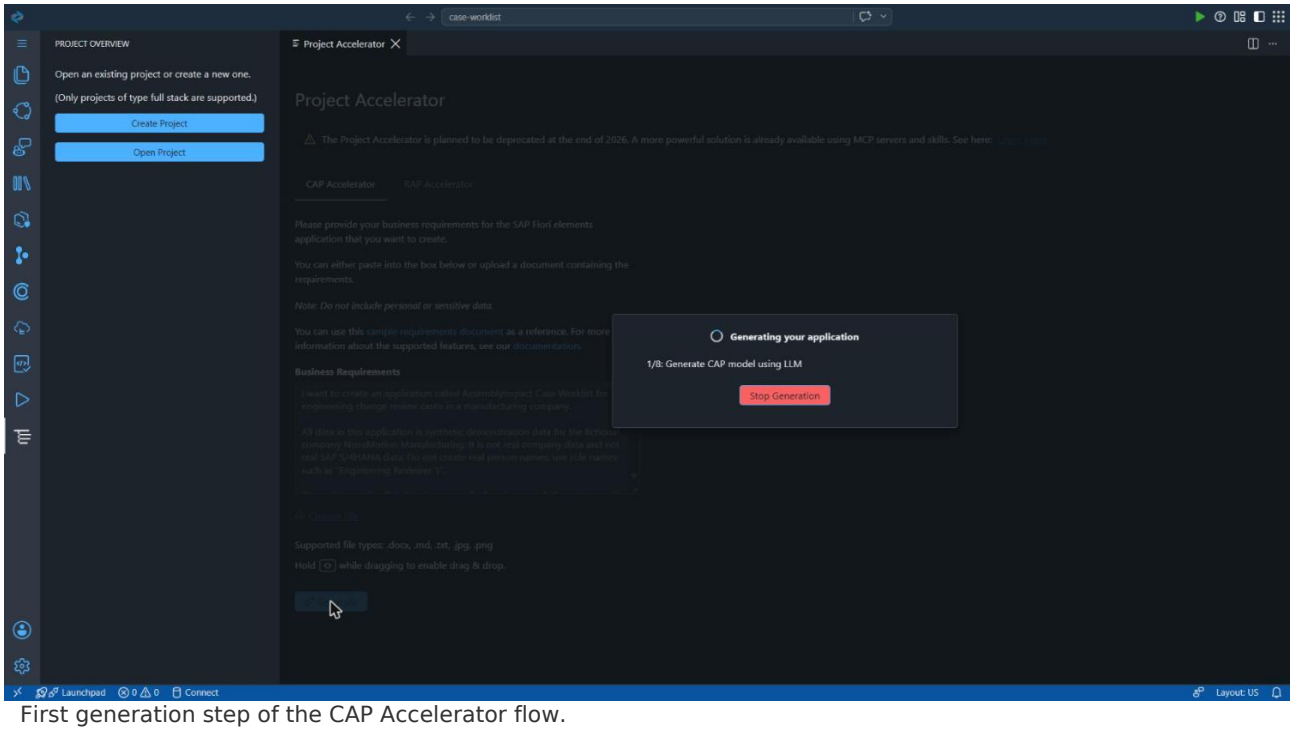
Repository in BAS fast-forwarded to main a0068c6 (M04 Slice 1b merge); baseline npm test 35/35 and lint pass before any P3 work.
Captured 2026-11-09T03:30:20Z UTC · file screenshots/P3-01_bas-synced-main-regression.jpg · SHA-256 c164cf555a7a44f2…

P3-02 — Project Accelerator input in the separate sandbox



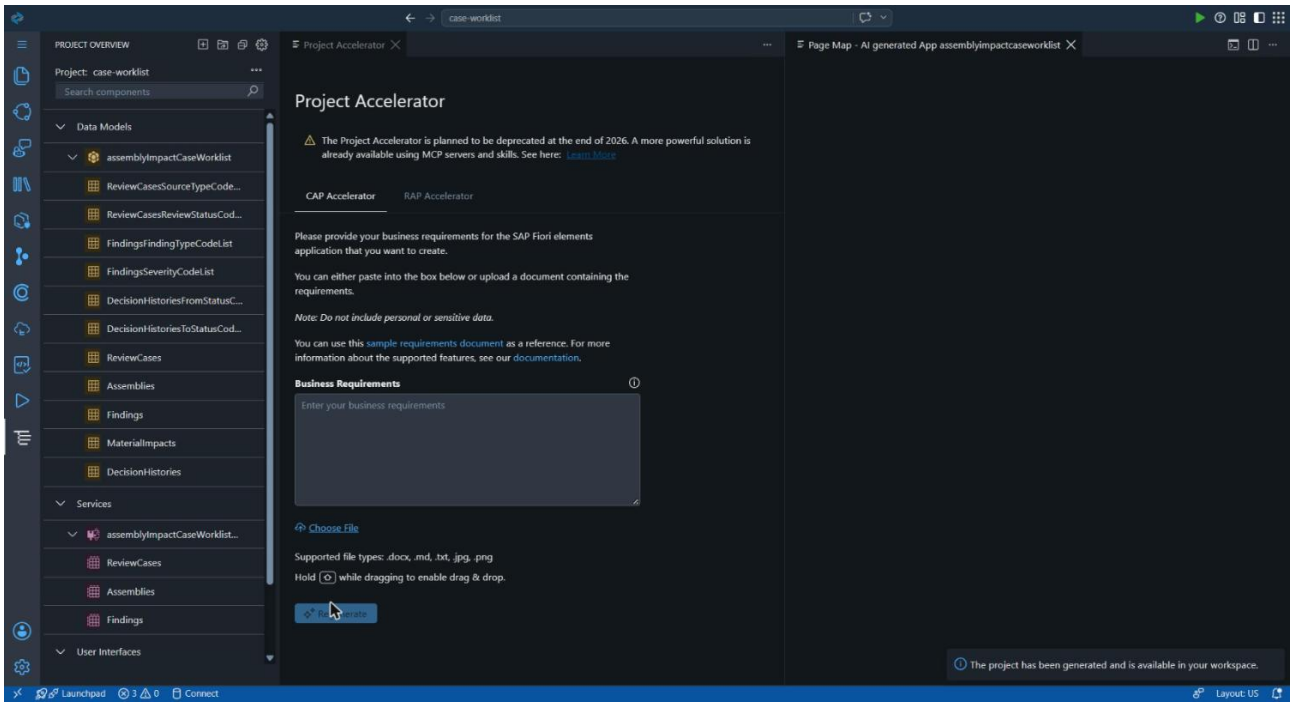
Project Accelerator opened in pa-sandbox/case-worklist with the requirements document and the approved layout image as input. Not the AssemblyImpact repository.
Captured 2026-11-09T03:33:25Z UTC · file screenshots/P3-02_pa-input-requirements-loaded.jpg · SHA-256 eb762e28071947c4...

P3-03 — Project Accelerator generation running



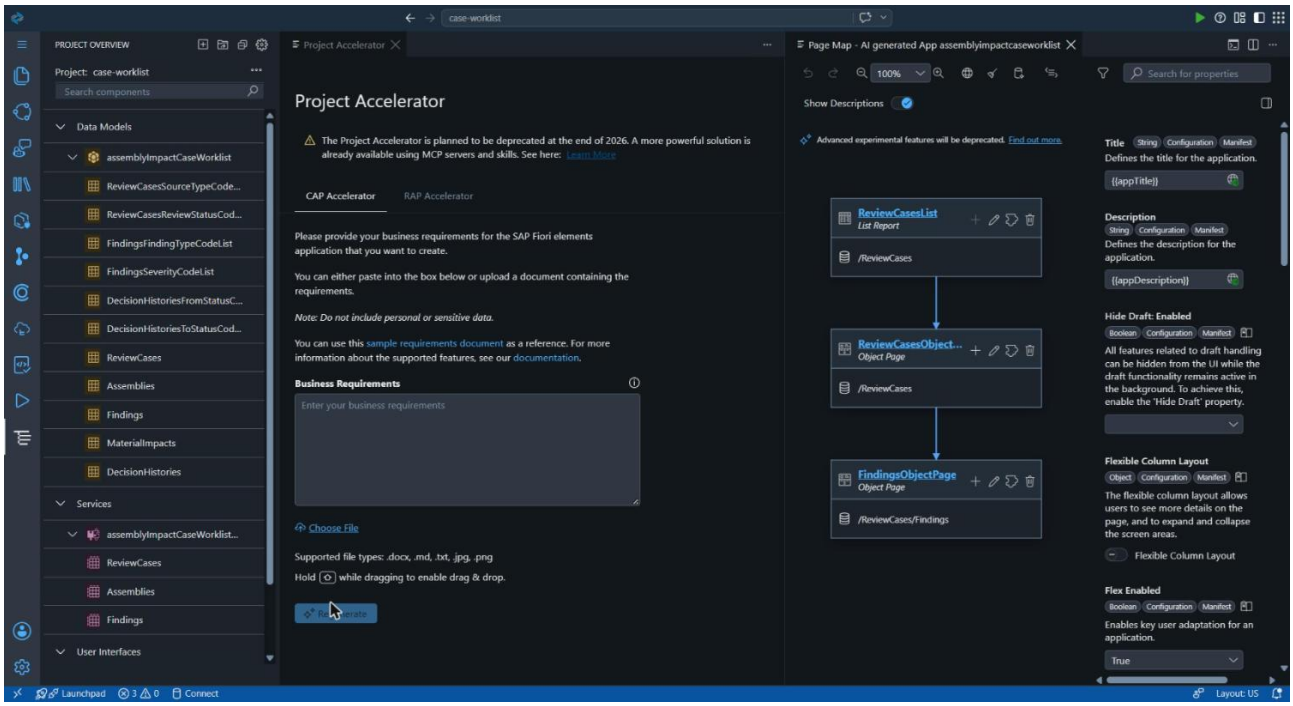
First generation step of the CAP Accelerator flow.
Captured 2026-11-09T03:33:41Z UTC · file screenshots/P3-03_pa-generation-step1.jpg · SHA-256 a419db4e584d288f...

P3-04 — Generation completed



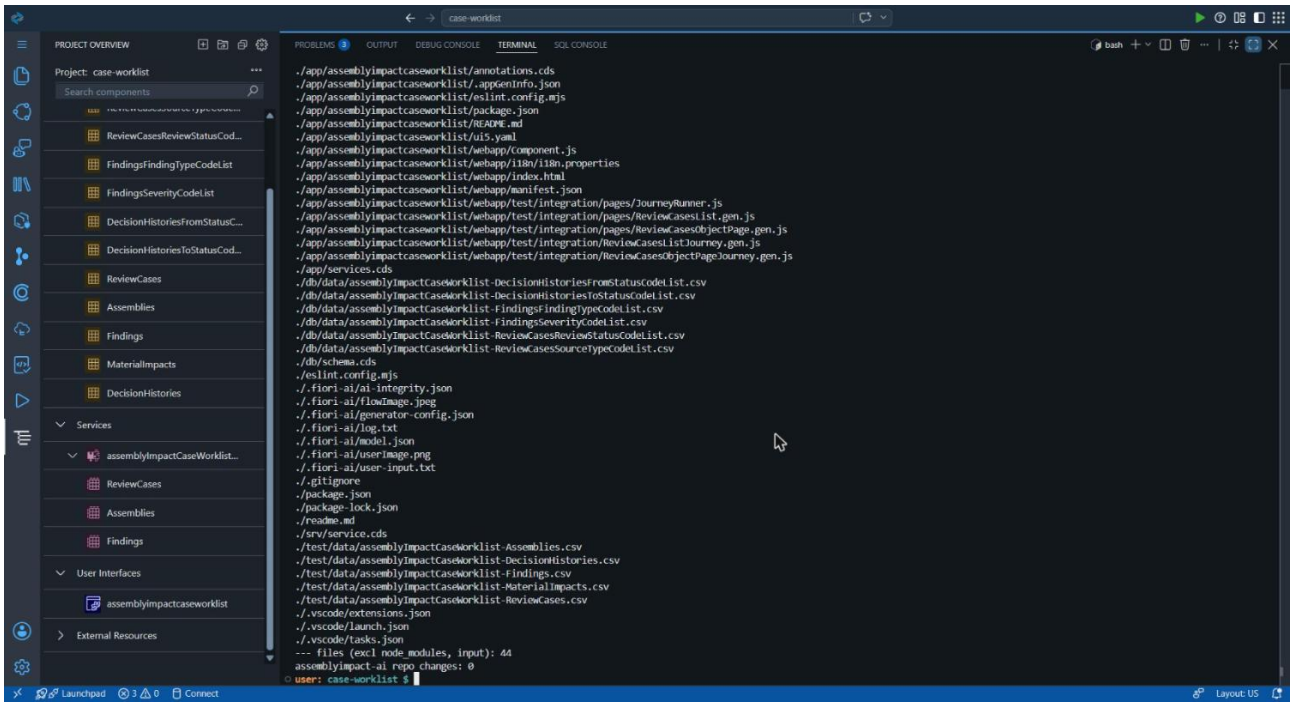
Project Accelerator reports the generated application assemblyImpactCaseWorklist.
Captured 2026-11-09T03:35:29Z UTC · file screenshots/P3-04_pa-generation-complete.jpg · SHA-256 765742603311cfce...

P3-05 — Generated Page Map (design-time view, not a running UI)



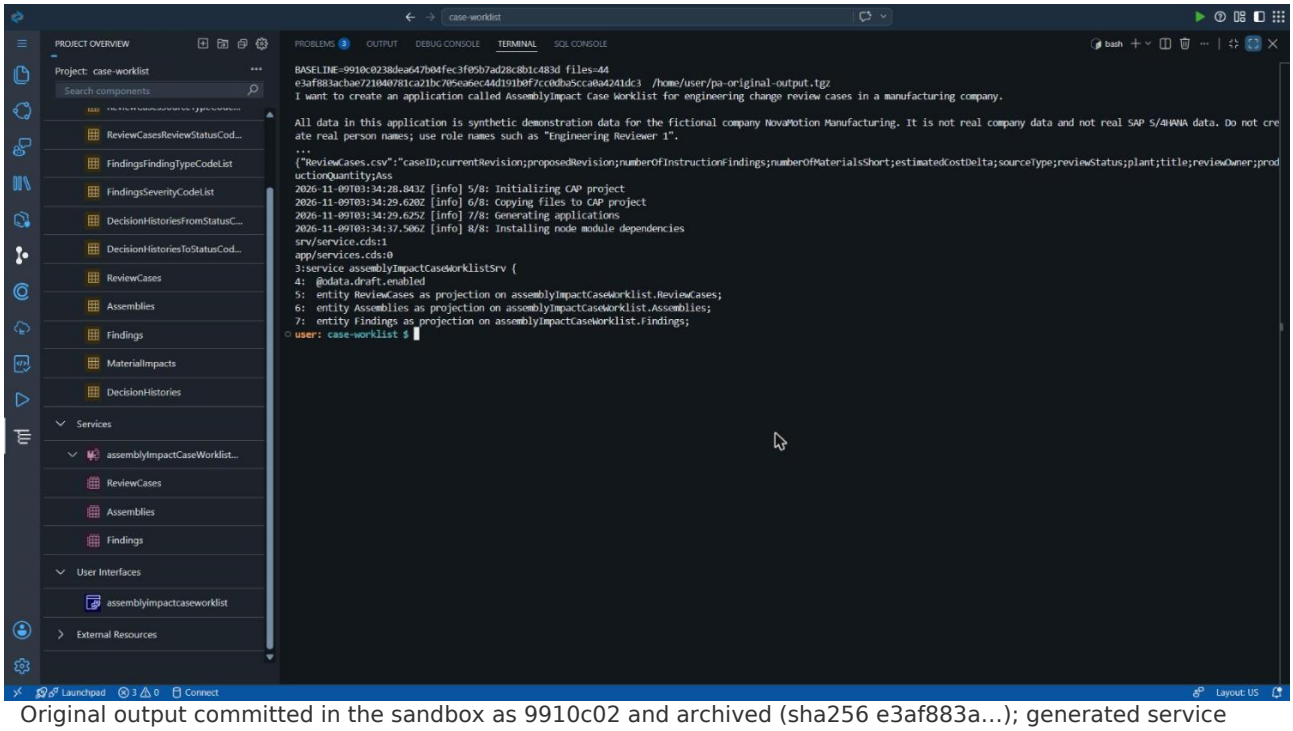
Page Map of the generated app: ReviewCases List Report -> ReviewCases Object Page -> Findings Object Page. A configuration view only; the running UI is shown in P3-08/P3-09.
Captured 2026-11-09T03:35:29Z UTC · file screenshots/P3-05_pa-page-map.jpg · SHA-256 f40b009d556e9fd3...

P3-06 — Generated files



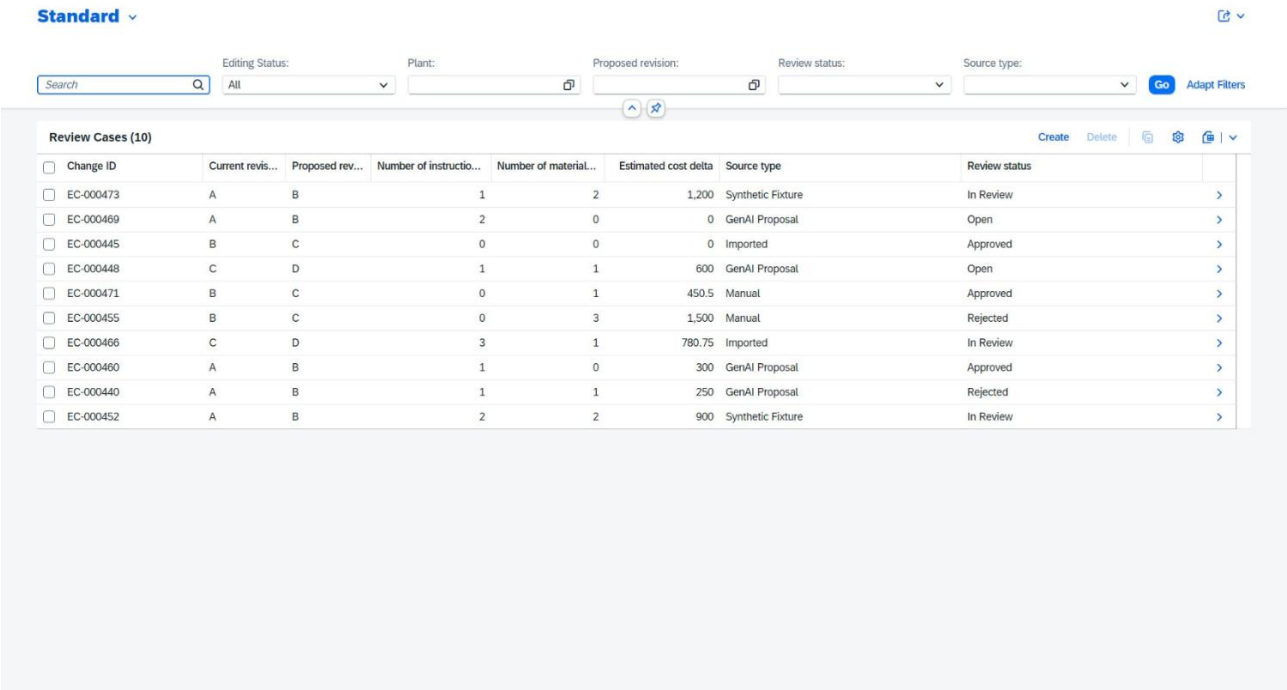
44 generated files: own CDS namespace, draft-enabled service, code lists, sample CSV data and a Fiori elements app.
Captured 2026-11-09T03:35:54Z UTC · file screenshots/P3-06_pa-generated-files.jpg · SHA-256 af9b66a143dab755...

P3-07 — Sandbox baseline commit and generated draft service



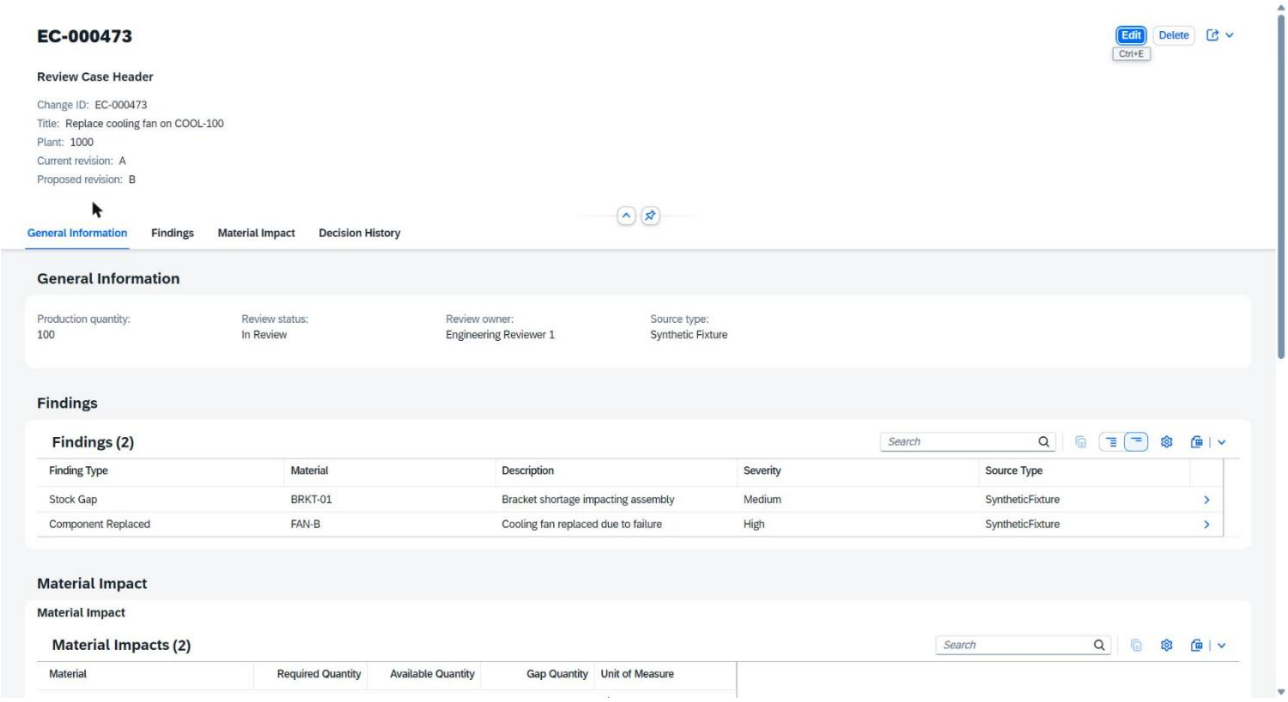
Original output committed in the sandbox as 9910c02 and archived (sha256 e3af883a…); generated service assemblyImpactCaseWorklistSrv is @odata.draft.enabled.
Captured 2026-11-09T03:36:27Z UTC · file screenshots/P3-07_pa-baseline-and-draft-service.jpg · SHA-256 021252bc97ca3293...

P3-08 — Generated UI running with its own mock data — List Report



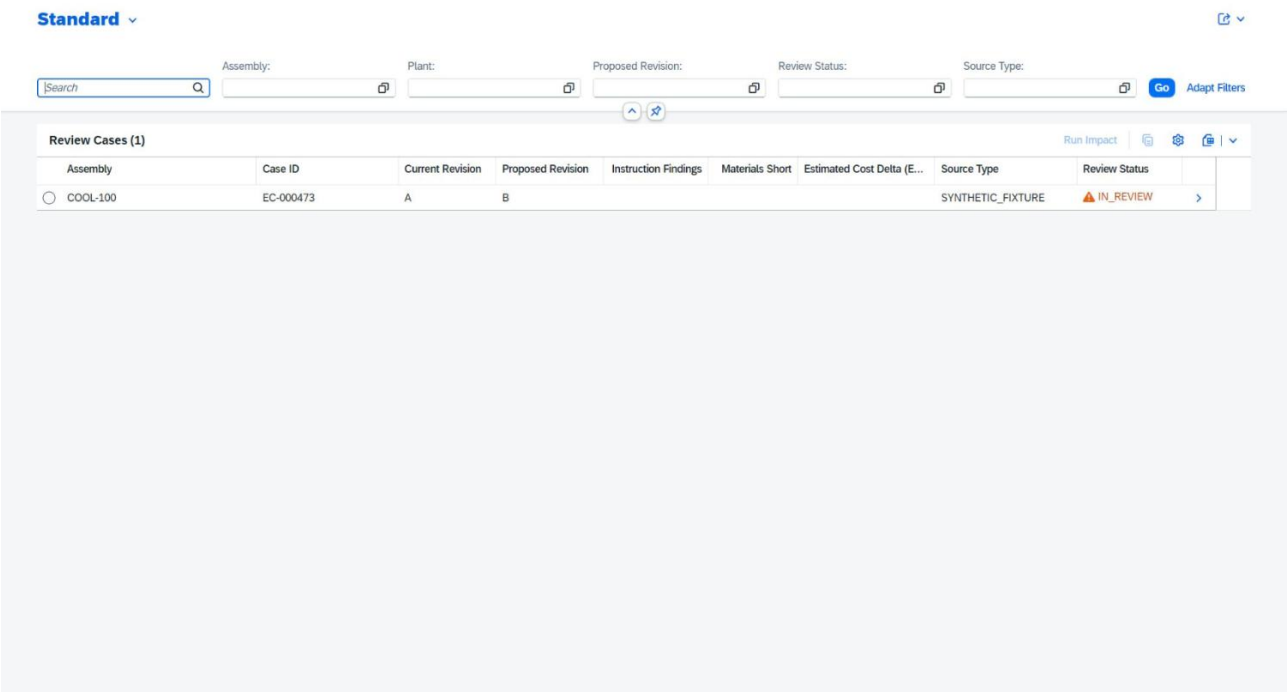
Browser-rendered generated app (sandbox cds watch). Shows 10 review cases invented by the generator, an Editing Status filter and Create/Delete buttons from draft handling. Not AssemblyImpact data.
Captured 2026-11-09T08:26:39Z UTC · file screenshots/P3-08_generated-ui-list-report.jpg · SHA-256 0d4091c6904e5b81…

P3-09 — Generated UI running with its own mock data — Object Page



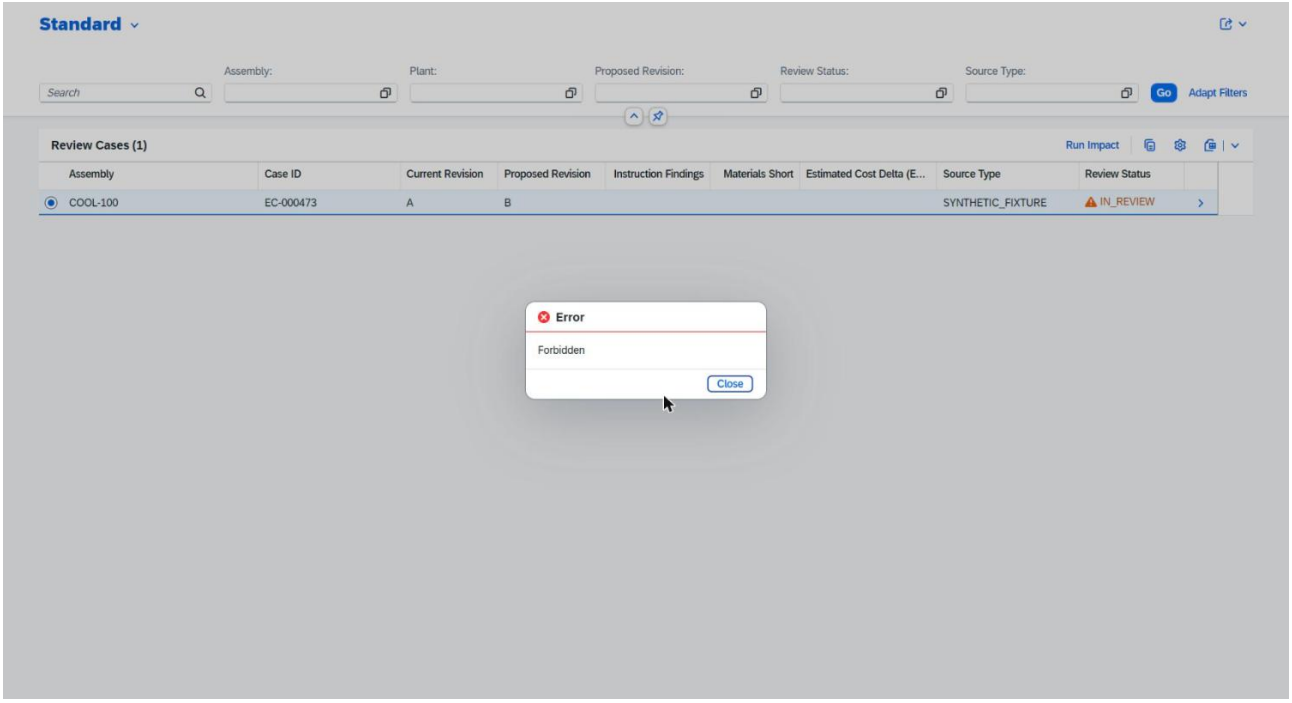
Generated Object Page with Edit/Delete and invented findings for EC-000473. Values do not match the backend oracle; this is why the generated data and service were not imported.
Captured 2026-11-09T08:27:00Z UTC · file screenshots/P3-09_generated-ui-object-page.jpg · SHA-256 7fb294cf9b5038fd…

P3-10 — Implemented worklist on EngineeringReview — viewer (plant 1000)



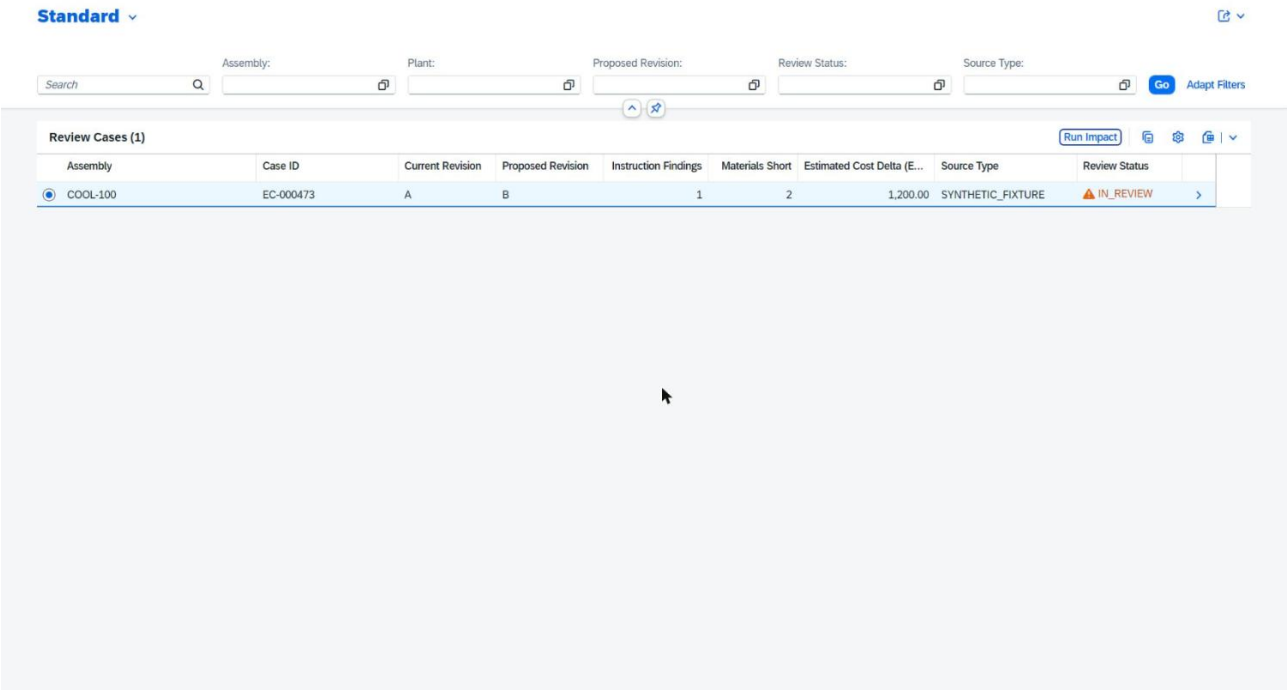
User cviewer sees one case, EC-000473 / COOL-100, IN_REVIEW. Impact columns are empty because no impact run exists yet.
Captured 2026-11-09T08:30:45Z UTC · file screenshots/P3-10_integrated-worklist-cviewer.jpg · SHA-256 c15bbd6685e7e1fb...

P3-11 — Run Impact refused for a viewer



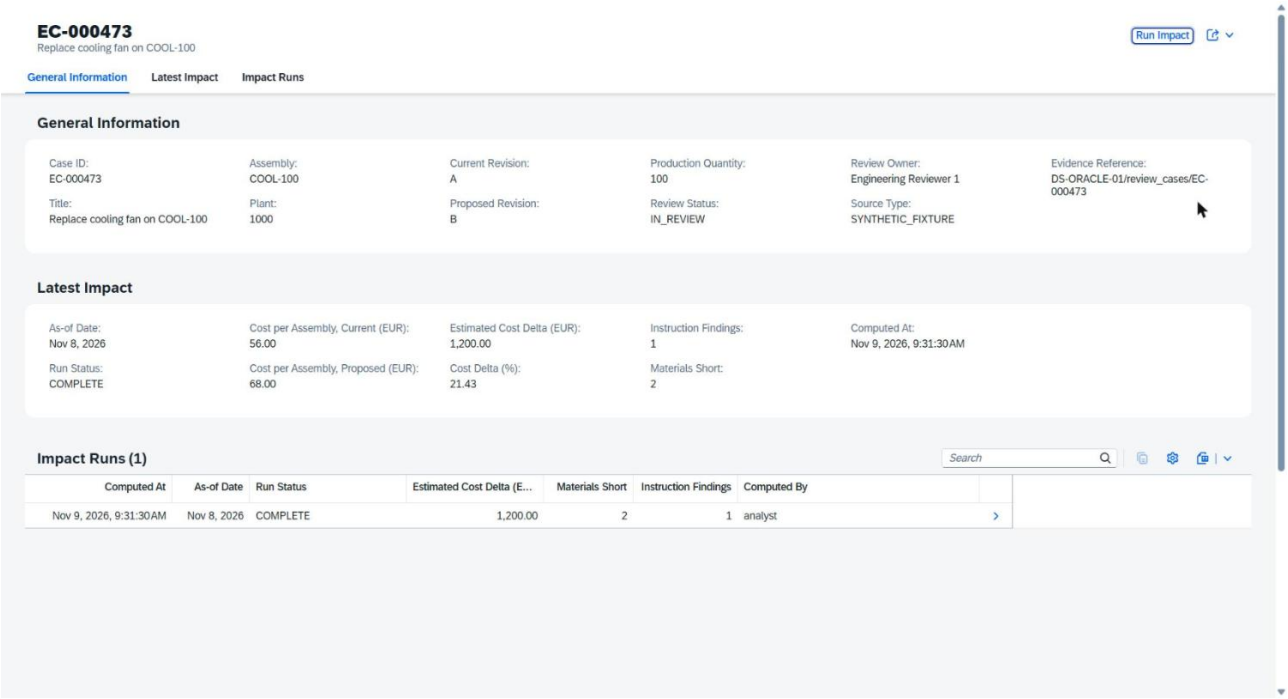
cviewer presses Run Impact; the server returns 403 Forbidden and the app shows the message. Authorization is enforced by the service.
Captured 2026-11-09T08:31:10Z UTC · file screenshots/P3-11_runimpact-forbidden-viewer.jpg · SHA-256 2d6e13b7d50e4807...

P3-12 — Run Impact by an analyst fills the worklist



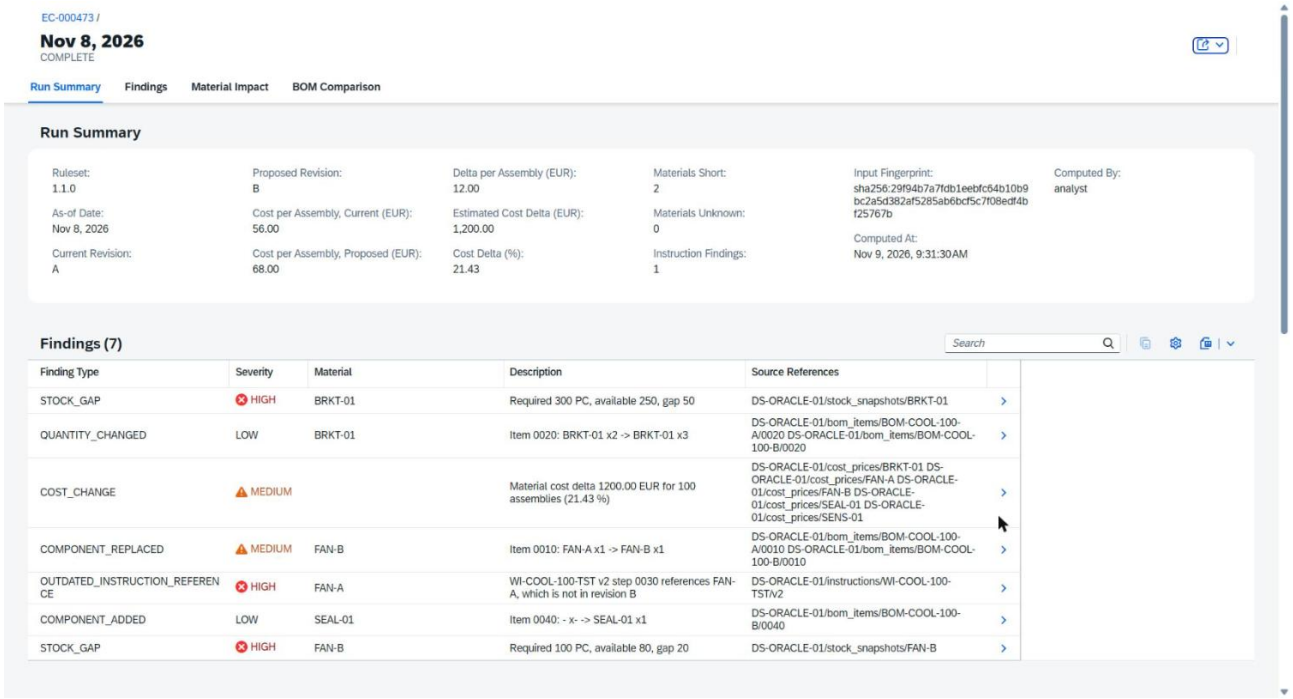
User analyst runs the impact for 2026-11-08: instruction findings 1, materials short 2, estimated cost delta 1,200.00 EUR (oracle values).
Captured 2026-11-09T08:31:44Z UTC · file screenshots/P3-12_runimpact-analyst-worklist-values.jpg · SHA-256 cf4bdeefcd2807eb...

P3-13 — Review Case Object Page



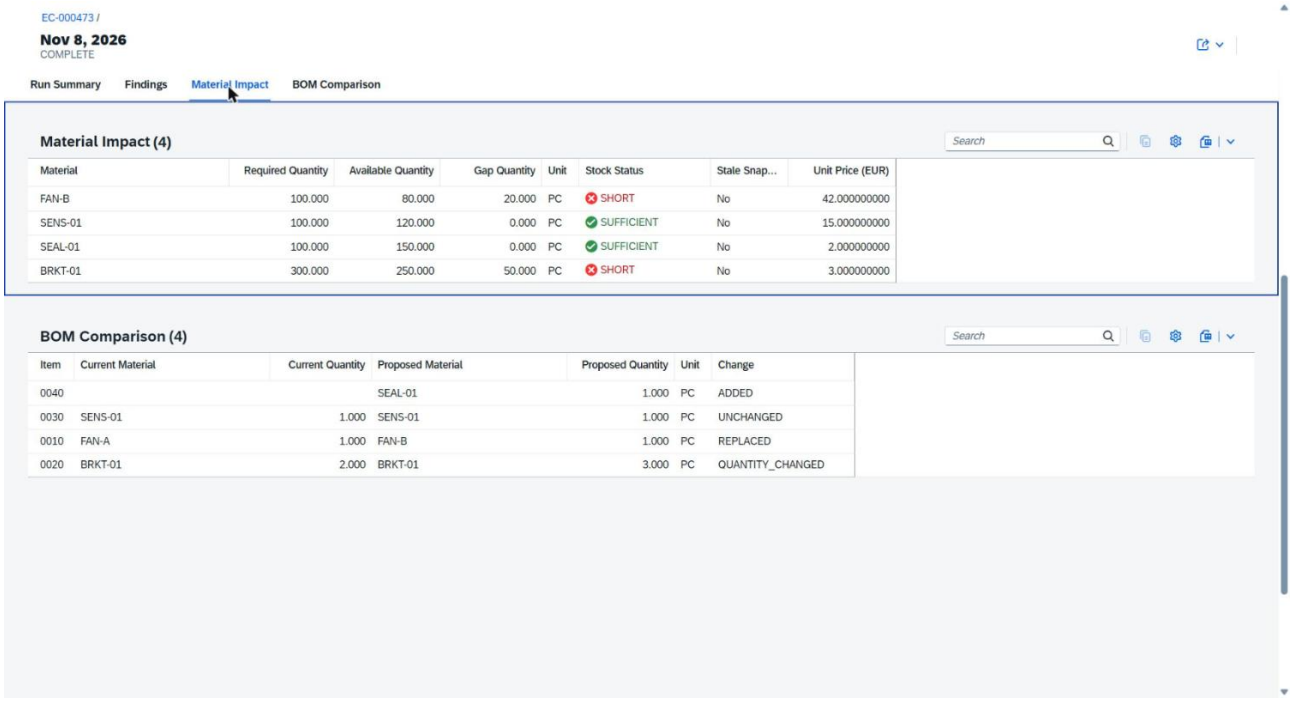
General section and Latest Impact (56.00 -> 68.00 EUR per assembly, +1,200.00 EUR, 21.43 %) and the list of impact runs.
Captured 2026-11-09T08:31:58Z UTC · file screenshots/P3-13_case-object-page.jpg · SHA-256 b775f9cca0469b3e...

P3-14 — Impact Run page — findings



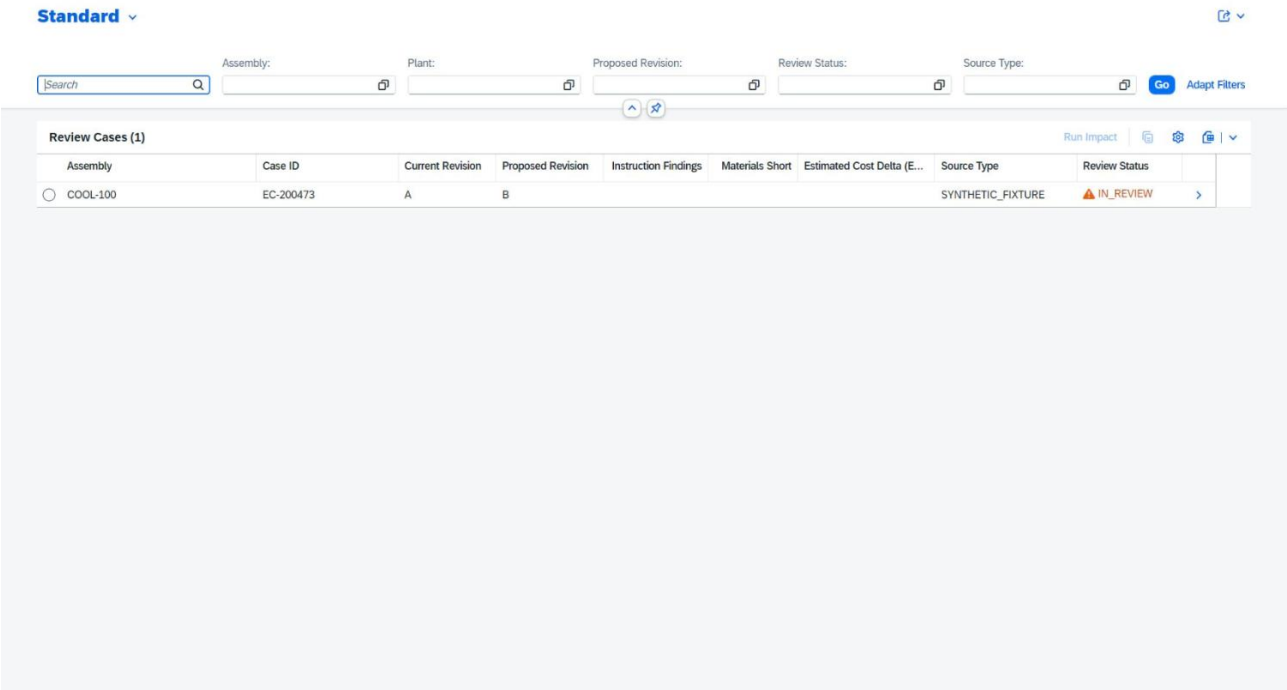
7 findings with severity criticality and evidence references. Captured before the sort-order fix (rows in backend order).
Captured 2026-11-09T08:32:13Z UTC · file screenshots/P3-14_impact-run-findings.jpg · SHA-256 88012287fdd0a5b6...

P3-15 — Impact Run page — material impact and BOM comparison



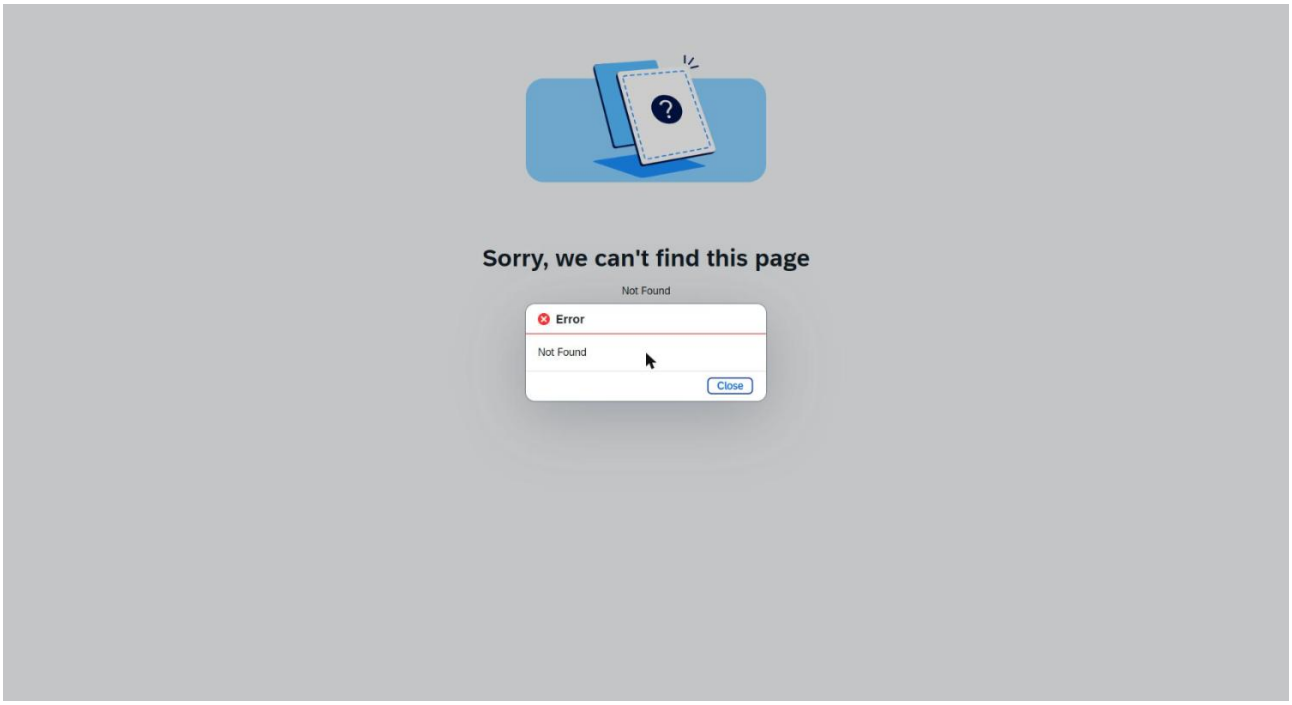
FAN-B gap 20 and BRKT-01 gap 50 marked SHORT; BOM comparison REPLACED / QUANTITY_CHANGED / UNCHANGED / ADDED. Captured before the sort-order fix.
Captured 2026-11-09T08:32:30Z UTC · file screenshots/P3-15_impact-run-materials-bom.jpg · SHA-256 8ad4dff1b9c61845...

P3-16 — Plant isolation in the UI



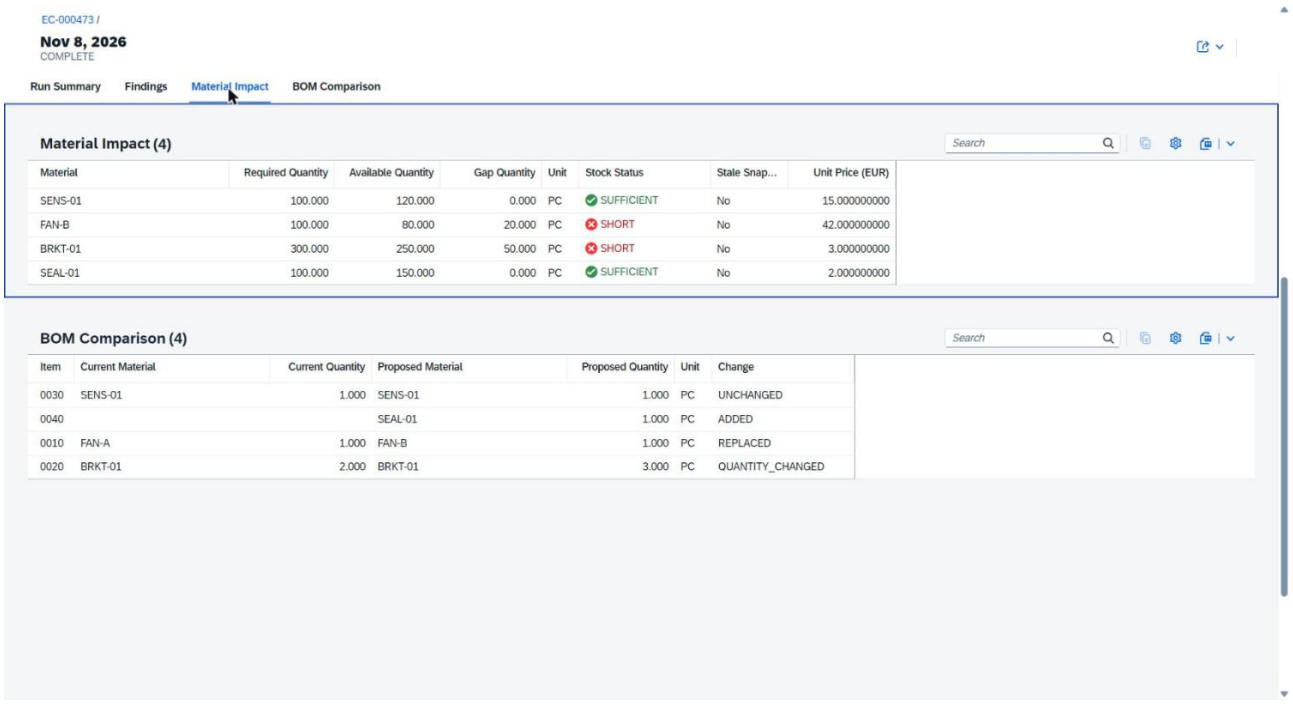
User analyst2000 (plant 2000) sees only the labelled mirror case EC-200473.
Captured 2026-11-09T08:32:47Z UTC · file screenshots/P3-16_plant2000-user-isolation.jpg · SHA-256 b767fbd2cee97df1…

P3-17 — Cross-plant deep link refused



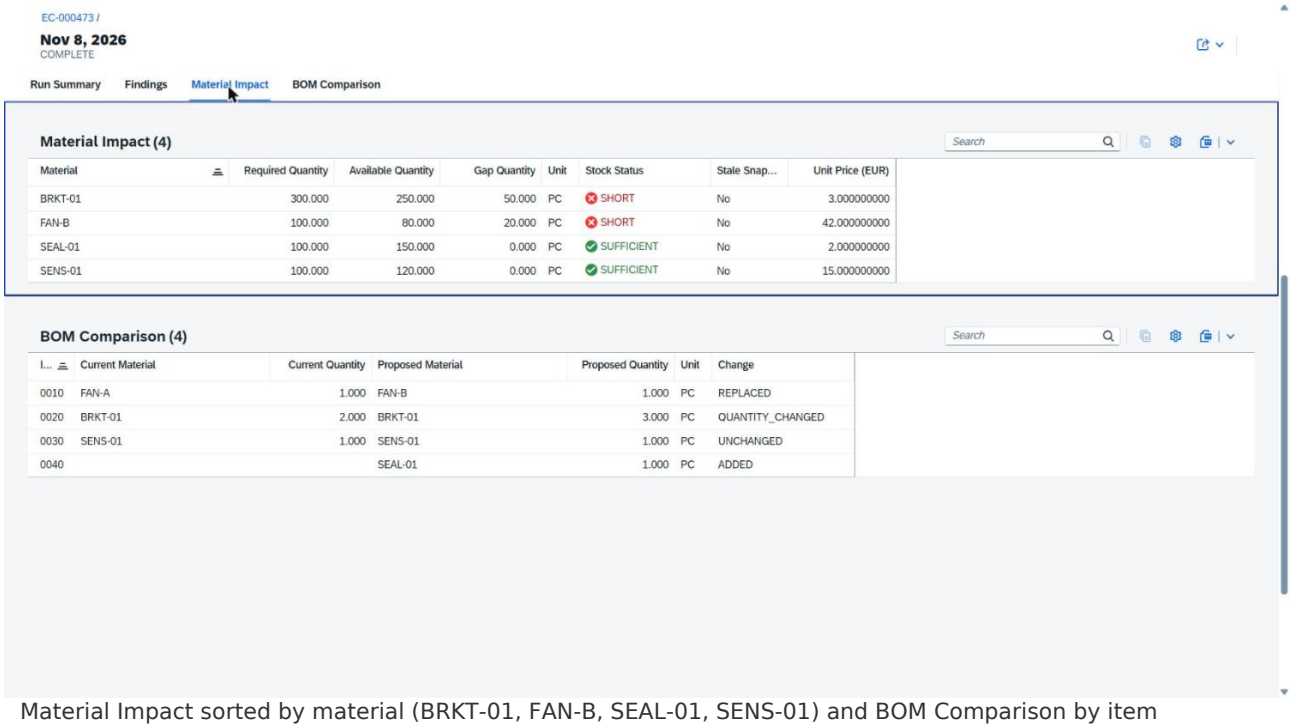
analyst2000 opens the plant-1000 case URL directly; the service answers 404 and the app shows Not Found.
Captured 2026-11-09T08:33:15Z UTC · file screenshots/P3-17_cross-plant-deeplink-not-found.jpg · SHA-256 0abfe1d864a38ea4…

P3-18 — Defect found in the browser: section tables not sorted



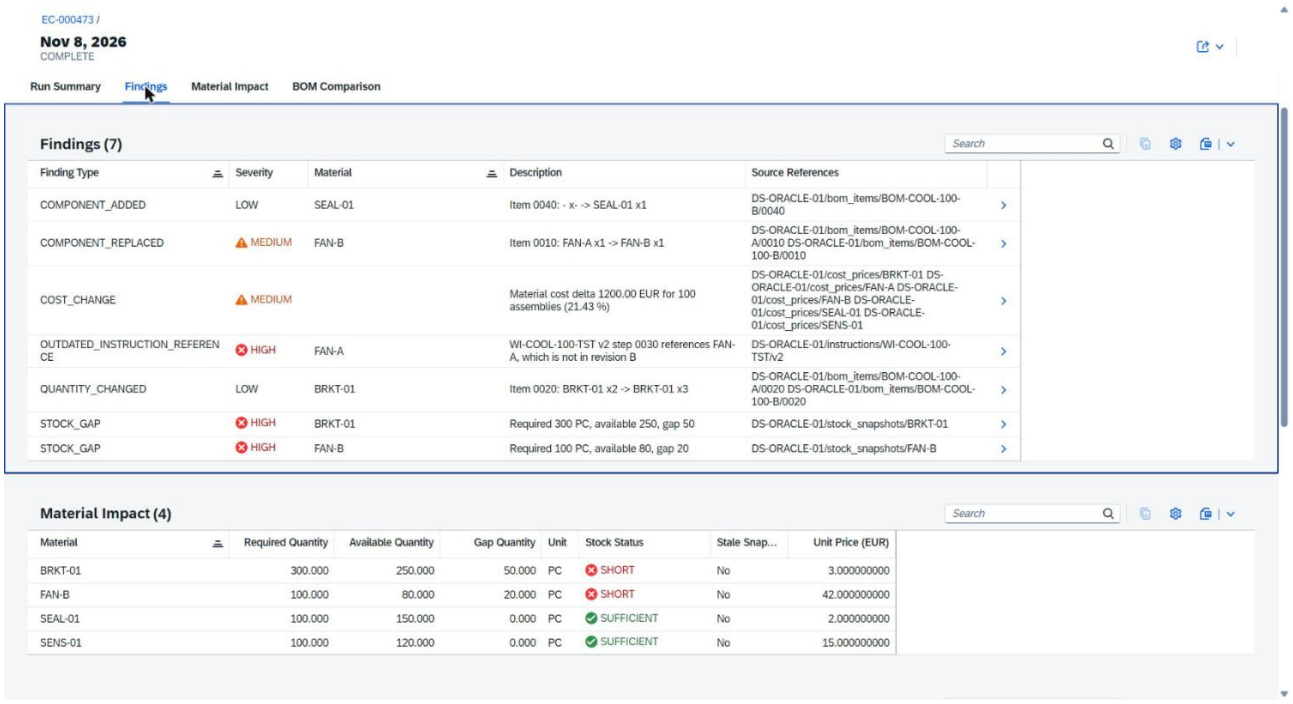
After commit af43aaf the Material Impact and BOM Comparison rows were still in insertion order (SENS-01 first, item 0030 first).
Captured 2026-11-09T08:37:25Z UTC · file screenshots/P3-18_before-fix-unsorted.jpg · SHA-256 70b0e016b53d2663...

P3-19 — After fix 377e46a: material and BOM tables sorted



Material Impact sorted by material (BRKT-01, FAN-B, SEAL-01, SENS-01) and BOM Comparison by item (0010-0040); sort indicators shown in the column headers.
Captured 2026-11-09T08:42:01Z UTC · file screenshots/P3-19_run-material-bom-sorted.jpg · SHA-256 55d82f82f3ca0e38...

P3-20 — After fix 377e46a: findings sorted by type and material



Findings sorted by finding type, then material; severity is shown with criticality icons.
Captured 2026-11-09T08:42:07Z UTC · file screenshots/P3-20_run-findings-sorted.jpg · SHA-256 48c0d6cd0e75c1cf...

9. Security review

Area	Result	Evidence
Authorization stays in the service	The UI adds no access logic; runImpact 403 for viewers and 404 across plants come from the backend	P3-11, P3-17, tests
Plant isolation	List, deep links and navigation paths respect the plant attribute	P3-16, P3-17, tests
Authentication	Anonymous OData request 401; the static app files are public but contain no data or secrets	tests
No write surface	No draft, create, update or delete; only the existing bound action	annotation test (no Draft)
Generated code	Generated service, data and draft handling not imported; sandbox kept separate	D-P3-1, D-P3-2
External resources	SAPUI5 loaded from sapui5.hana.ondemand.com 1.149.0 (SAP CDN); no other third-party script	index.html
Dependencies / secrets	No new packages; diff contains no keys or tokens	review

Unresolved risks. No xs-security.json or role templates (blocks BTP deployment); the Run Impact button is visible to viewers (server refuses); development mock passwords are public by design; the CDN dependency needs a CSP/approuter decision before deployment.

Review disclosure. Code review, security review and quality assessment were performed by one AI system (Claude, Anthropic). No human or SAP reviewer took part.

10. Limitations

- Unit price shows 9 decimals (backend Decimal scale); cosmetic, not fixed.
- Decision History, Create Change and Export from the mockup are not implemented (no backend support yet).
- Findings are not ordered by severity rank (string field); criticality icons mark severity.
- Run Impact uses a free date parameter dialog; no value help.

- Preview only in BAS with mocked authentication; no BTP deployment, approuter or launchpad tile.
- All data synthetic (NovaMotion Manufacturing); no S/4HANA connection; no Joule.

11. GitHub traceability

Item	Value	Status
Branch	feat/p3-case-worklist (from main a0068c6)	verified
Code commits	77730307a1ab; af43aafe863e; 377e46aa5b71 (head 377e46aa5b71)	verified in BAS (same hashes)
Changed files (code)	10 files: app/case-worklist (6), test/case-worklist-app.test.js, package.json, readme.md	verified
Evidence commit	docs(p3): add Case Worklist evidence (this folder); a file cannot contain its own commit hash	see PR
Pull request	not created	pending
CI	GitHub Actions: not yet available when this report was generated	pending
main	unchanged at a0068c6050ca	verified before push

12. Four-project relevance and next steps

Project	Contribution	Not claimed
P1 Joule	Same authorized service a skill could call.	No Joule skill.
P2 S/4HANA + GenAI	UI ready to show S4_IMPORT cases later.	No S/4HANA data.
P3 Project Accelerator	Generated, run, compared and adapted; evidence of every correction.	Page Map is not a UI.
P4 custom investigation UI	None (kept separate).	Not started.

- Owner review of D-P3-1 to D-P3-8 and of this PR; merge only with separate approval.
- Approval-workflow slice: decision history entity and actions, then the Decision History section.
- BTP readiness: xs-security.json, approuter, CSP for the UI5 CDN, HANA; deployment needs separate authorization.

Appendix A — Evidence manifest (screenshots)

ID	File	Captured (UTC)	SHA-256
P3-01	P3-01_bas-synced-main-regression.jpg	2026-11-09T03:30:20Z	c164cf555a7a44f24fc90e0a4c9c5884a2bb5d28db129ab4be4141d024969d9f
P3-02	P3-02_pa-input-requirements-loaded.jpg	2026-11-09T03:33:25Z	eb762e28071947c4b6f41314c951fdbbb8acb604db3834cd0ffa467405ca5357
P3-03	P3-03_pa-generation-step1.jpg	2026-11-09T03:33:41Z	a419db4e584d288f37358e0c4e34b2a8665473d3e6017908d96da3c73423ad4a
P3-04	P3-04_pa-generation-complete.jpg	2026-11-09T03:35:29Z	765742603311cfceb07aef8ce3310a78442bdb4dc5b11c3b9b74ca83f084150b
P3-05	P3-05_pa-page-map.jpg	2026-11-09T03:35:29Z	f40b009d556e9fd3f6cab163483ac08ebc910fff619ea445146f13131f4002b6
P3-06	P3-06_pa-generated-files.jpg	2026-11-09T03:35:54Z	af9b66a143dab755f236f7a27f2d4bf6a28c344c38aac971fd22f217c64f46dd
P3-07	P3-07_pa-baseline-and-draft-service.jpg	2026-11-09T03:36:27Z	021252bc97ca329356c5d29e0e1eaa87337c3d39d5a5adf5bb615c1028de887c
P3-08	P3-08_generated-ui-list-report.jpg	2026-11-09T08:26:39Z	0d4091c6904e5b812d19ae88cdc4fae32c8282213723c66cafaf805ce9148683
P3-09	P3-09_generated-ui-object-page.jpg	2026-11-09T08:27:00Z	7fb294cf9b5038fdce11355876757c2a25174b36c5cf77e4d8bc1c3caab178da
P3-10	P3-10_integrated-worklist-cviewer.jpg	2026-11-09T08:30:45Z	c15bbd6685e7e1fb2ae864a1d17439fa0a906f7e5b10d07dba03950055f66fd1
P3-11	P3-11_runimpact-forbidden-viewer.jpg	2026-11-09T08:31:10Z	2d6e13b7d50e4807eae1f0ba9ab6ff63d7749327dca7f42033603359a9340420
P3-12	P3-12_runimpact-analyst-worklist-values.jpg	2026-11-09T08:31:44Z	cf4bdeefcd2807eb168cb2d56cb6d42b069e482e9f6d584518d203f19f197ad4
P3-13	P3-13_case-object-page.jpg	2026-11-09T08:31:58Z	b775f9cca0469b3e1826994001dab26da4817a0b990e0a65369033c72bb1d5fa
P3-14	P3-14_impact-run-findings.jpg	2026-11-09T08:32:13Z	88012287fdd0a5b607d133bcdb8d812d556c339ca912b44777d861fa01384148
P3-15	P3-15_impact-run-materials-bom.jpg	2026-11-09T08:32:30Z	8ad4dff1b9c6184554569432f9e3b83294bf7bb37914716ae6795caebcd1f19a
P3-16	P3-16_plant2000-user-isolation.jpg	2026-11-09T08:32:47Z	b767fbd2cee97df12dbc760e426f965deb5b5cb6182629d0f4d78dabf51c2ee9
P3-17	P3-17_cross-plant-deeplink-not-found.jpg	2026-11-09T08:33:15Z	0abfe1d864a38ea4e2f4a50024178480608b6eedd128f3ec2dcb0b2bab082759
P3-18	P3-18_before-fix-unsorted.jpg	2026-11-09T08:37:25Z	70b0e016b53d2663a9a7a1e86a43f0f15d661ac4ab47c39c1c2b3d48fd0c9a96
P3-19	P3-19_run-material-bom-sorted.jpg	2026-11-09T08:42:01Z	55d82f82f3ca0e387c3e649aa7d29294fe331cf54525704b8470d5d0b2870061
P3-20	P3-20_run-findings-sorted.jpg	2026-11-09T08:42:07Z	48c0d6cd0e75c1cfc540f8a28e248c48c9508e24a85460bb78d92e06c14cd5bd

All files in this folder, including generated/ and logs/, are hashed in evidence-manifest.json, produced in BAS when the evidence was committed.

Appendix B — References

- SAP Fiori elements for OData V4: table sections and presentation variants — https://ui5.sap.com/#/topic/7f844f1021cd4791b8f7408eac7c1cec
- SAP CAP: serving Fiori UIs — https://cap.cloud.sap/docs/advanced/fiori
- Project artifacts: srv/engineering-review.cds, test/data (SEED-M04-01), docs/evidence/m04-slice1b.
- AI assistance: implementation, tests, evidence capture and this report by Claude (Anthropic, Claude Opus 5.5) under owner authorization; student review pending.